

Designation

디지털제어 및 신호처리 - 2조

유민결 유채은 김솔

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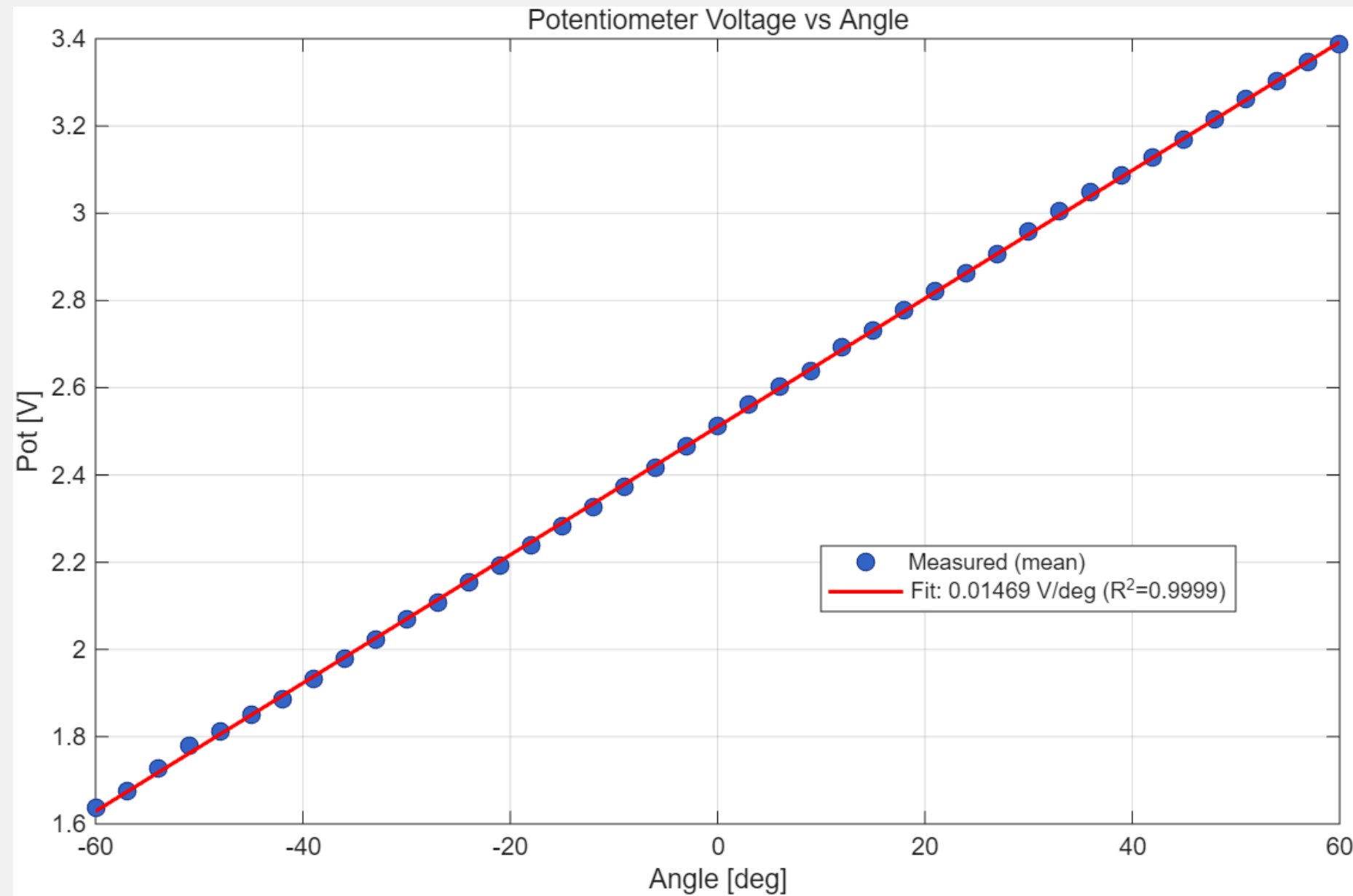
01 센서 모델링

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03 제어기 설계

04 성능분석

01 센서 모델링



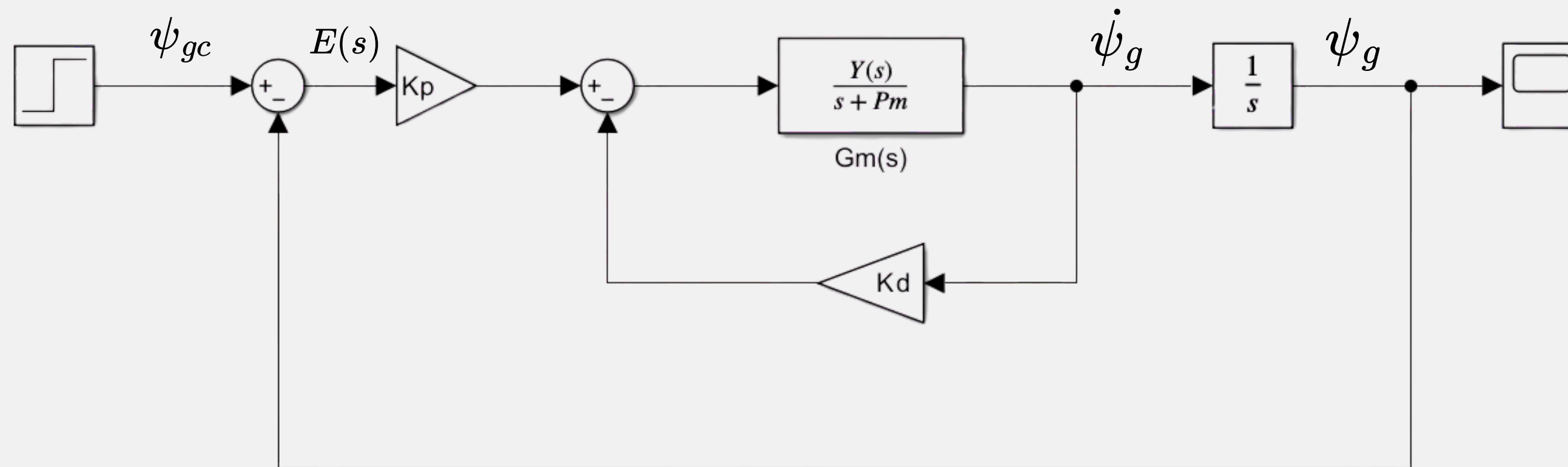
$$\psi_g = K_{pot}(V_{pot} - V_{neutral})$$

$$K_{pot} \approx 68.07352 [deg/V]$$

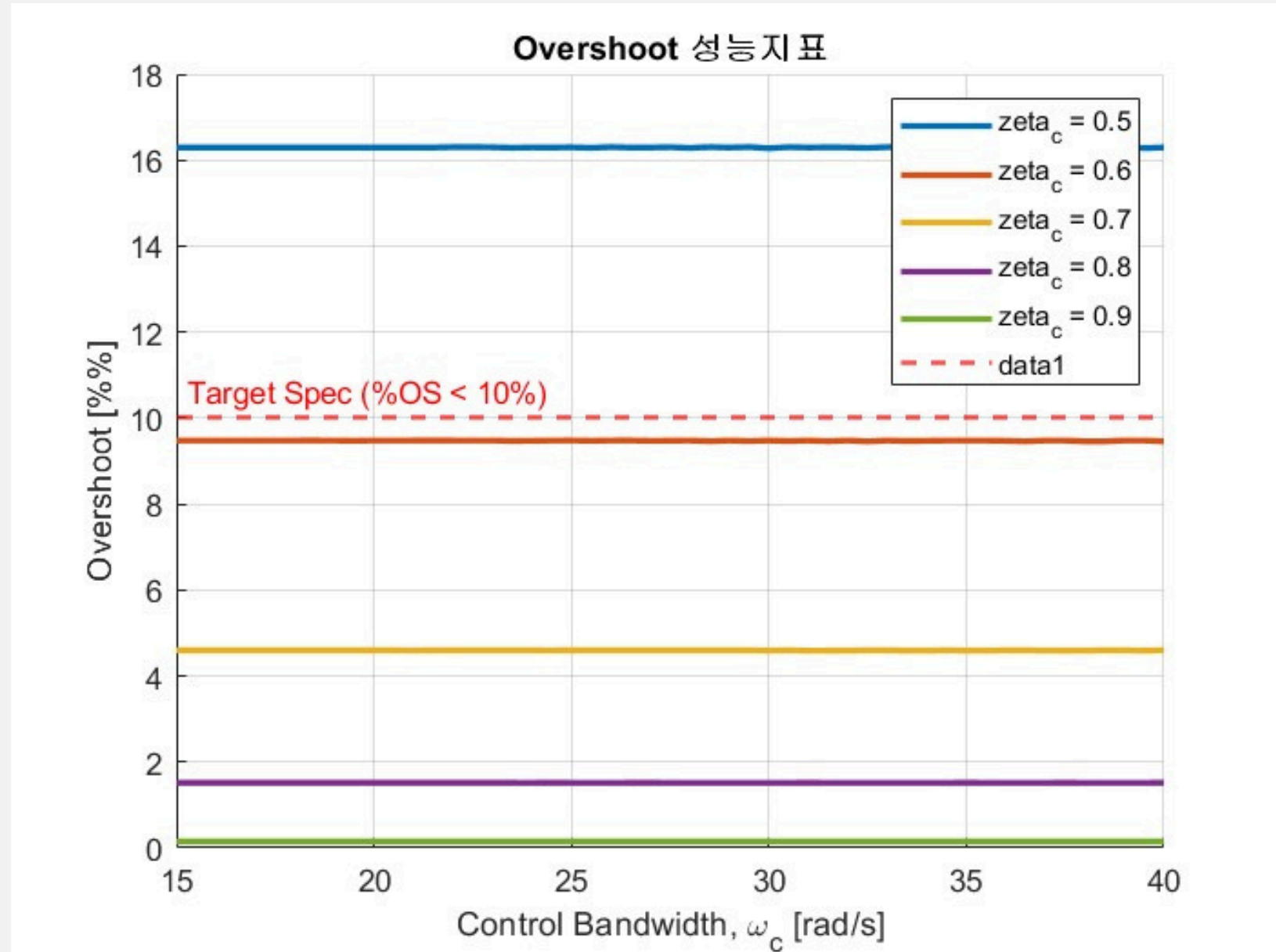
$$0[deg]에서 V_{neutral} = 2.5 [V]$$

02 제어기 구조

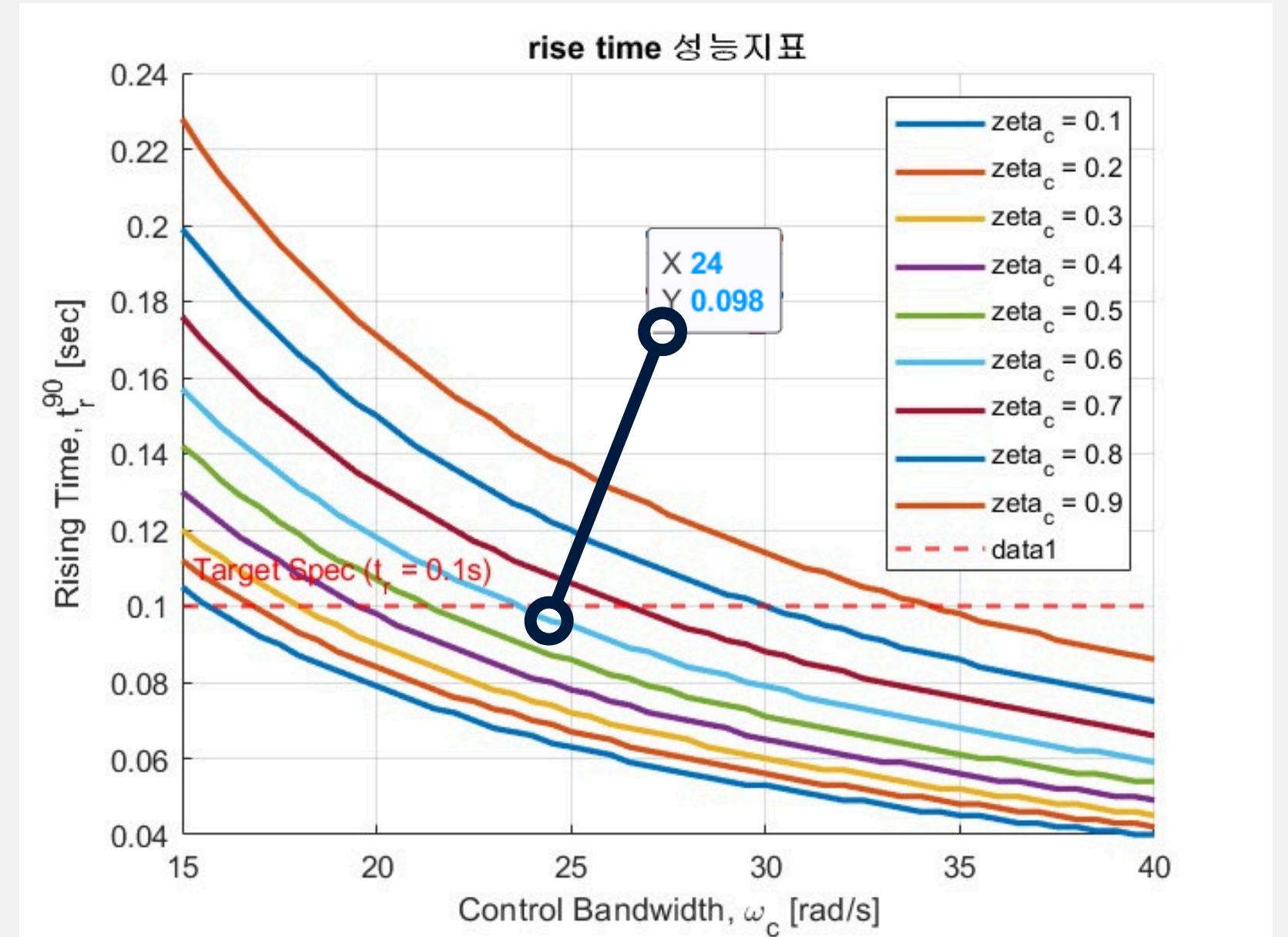
PD-controller



03 제어기 설계



$$\zeta_c \geq 0.6$$



$$\omega_c \geq 24$$

03 제어기 설계

$$G_m(s) = \frac{K_m}{s + P_m} = \frac{9.993}{s + 10.87}$$

$$G_{cl}(s) = \frac{K_P K_m}{s^2 + (P_m + K_D K_m)s + K_P K_m}$$

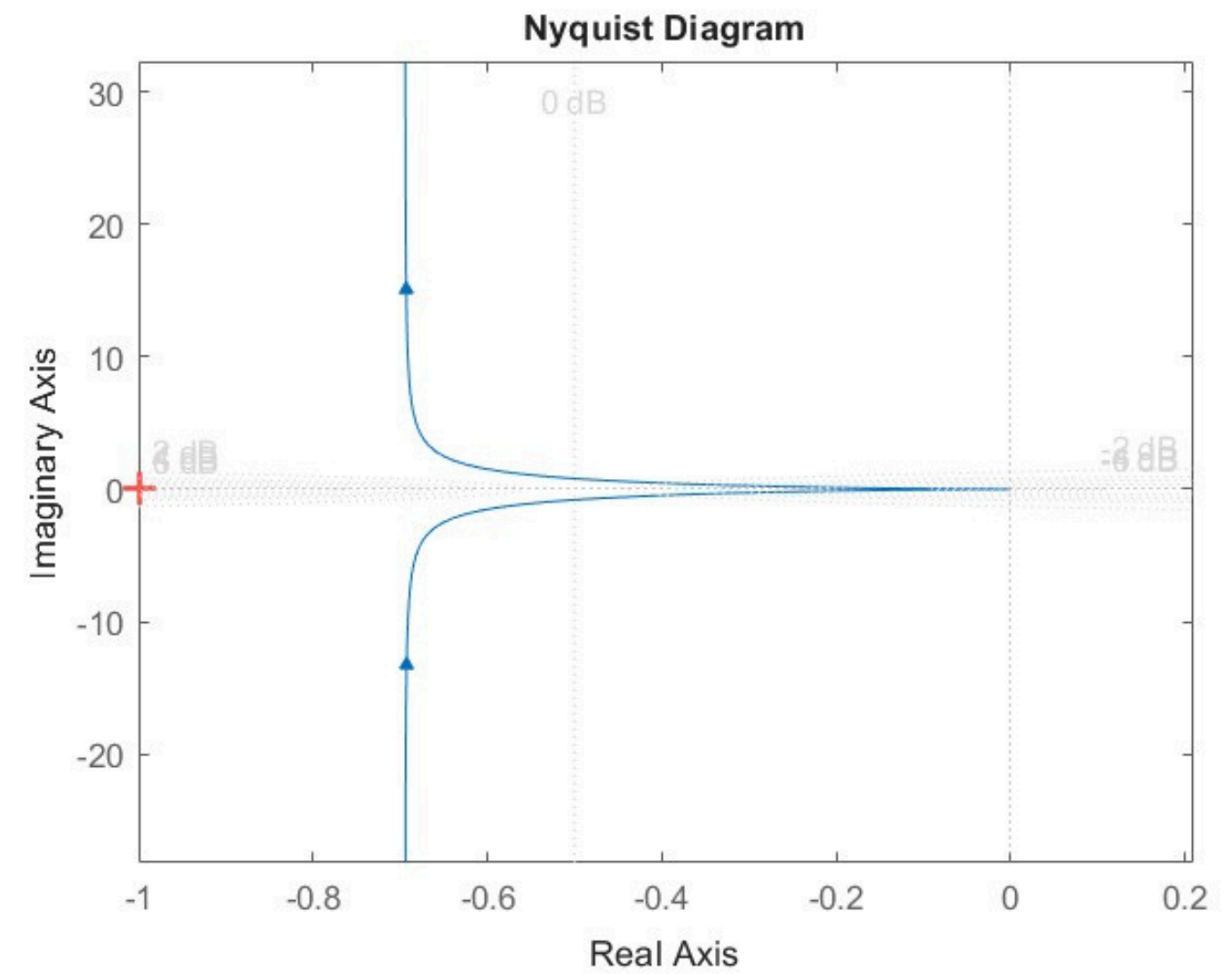
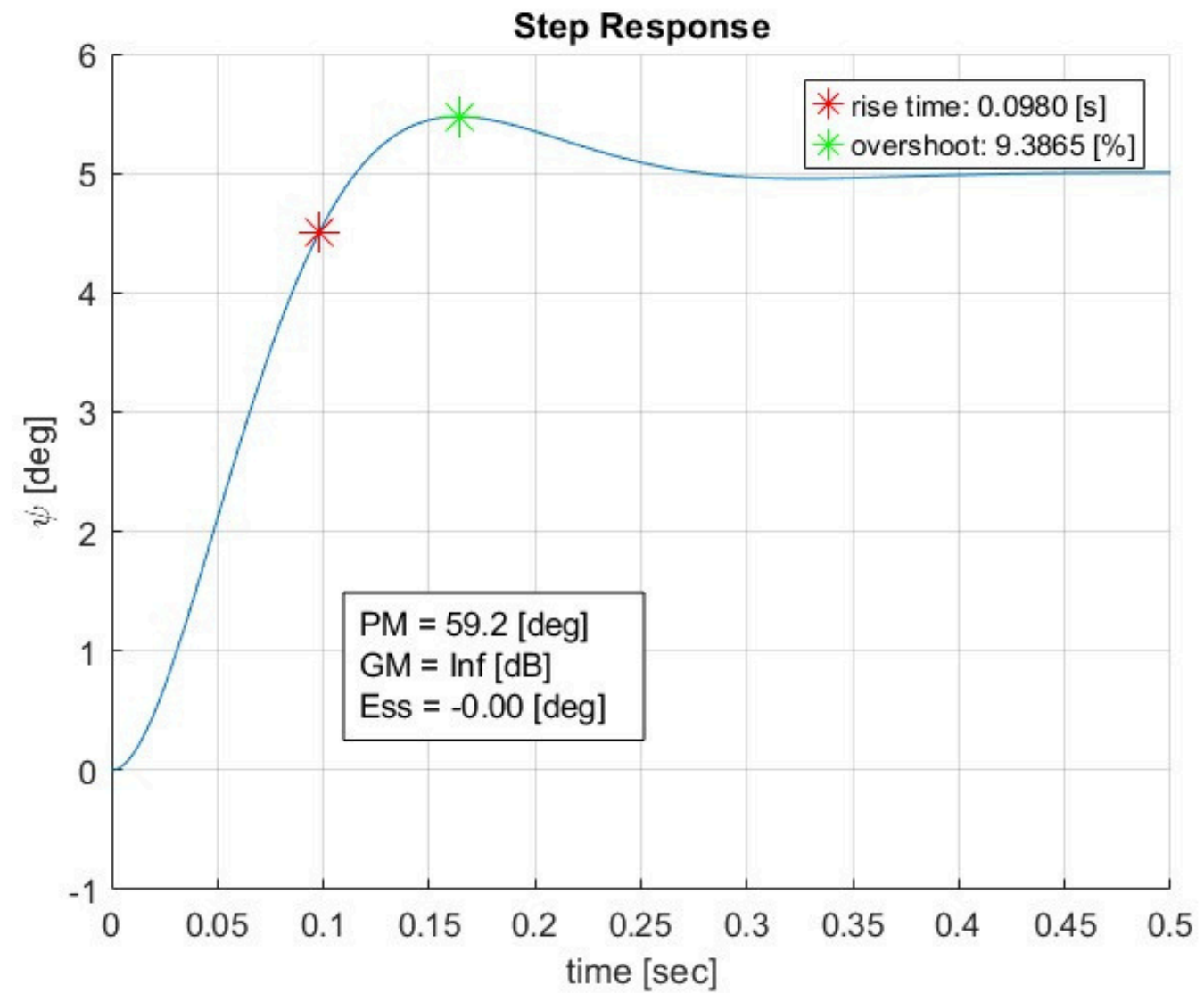
$$K_P = \frac{\omega_c^2}{K_m}, \quad K_D = \frac{2\zeta_c \omega_c - P_m}{K_m}$$

$$\omega_c = 24[rad/s] \rightarrow K_P \approx 57.6403 [1/s]$$

$$\zeta_c = 0.6[-] \rightarrow K_D \approx 1.7943$$

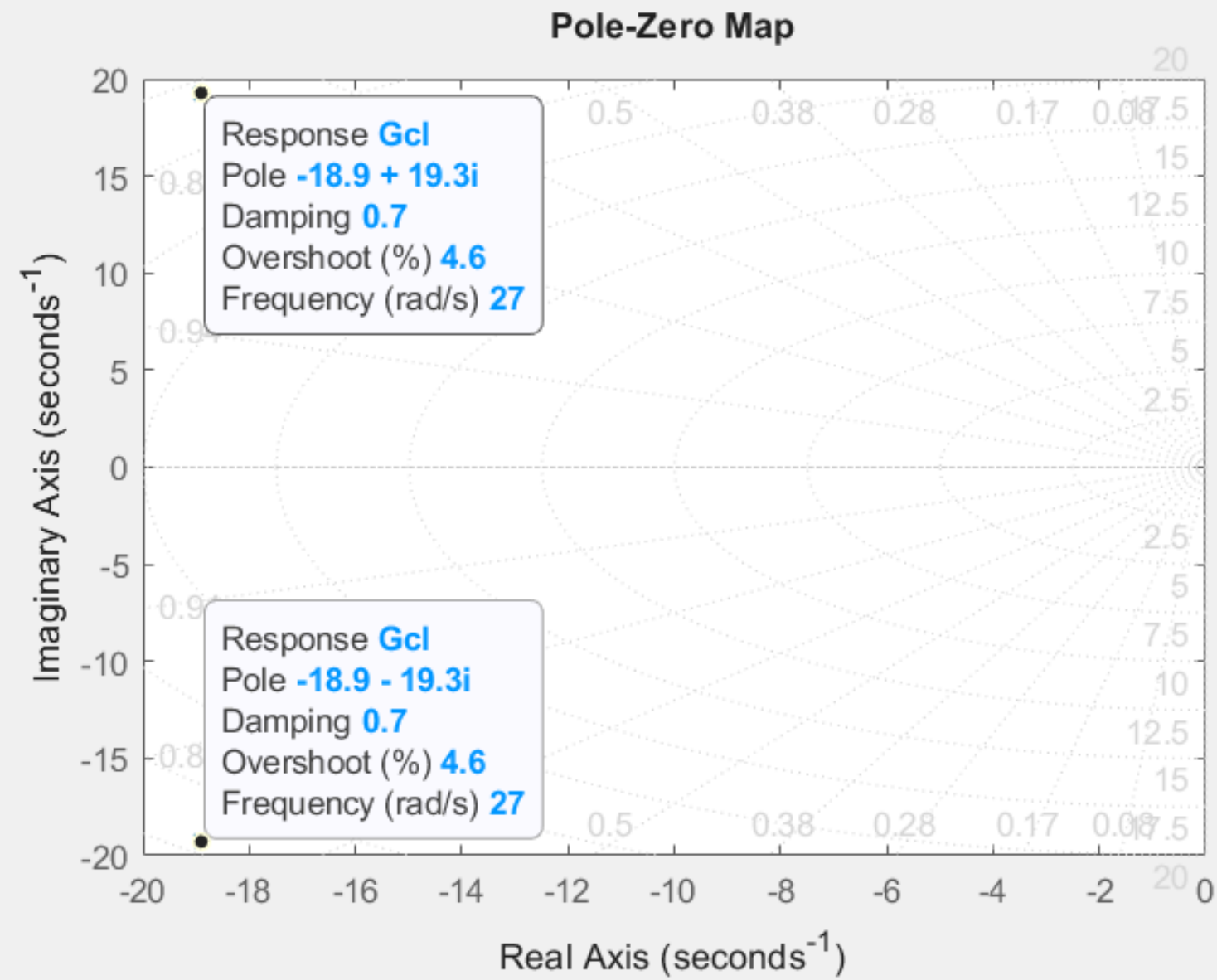
04 성능분석

설계모델



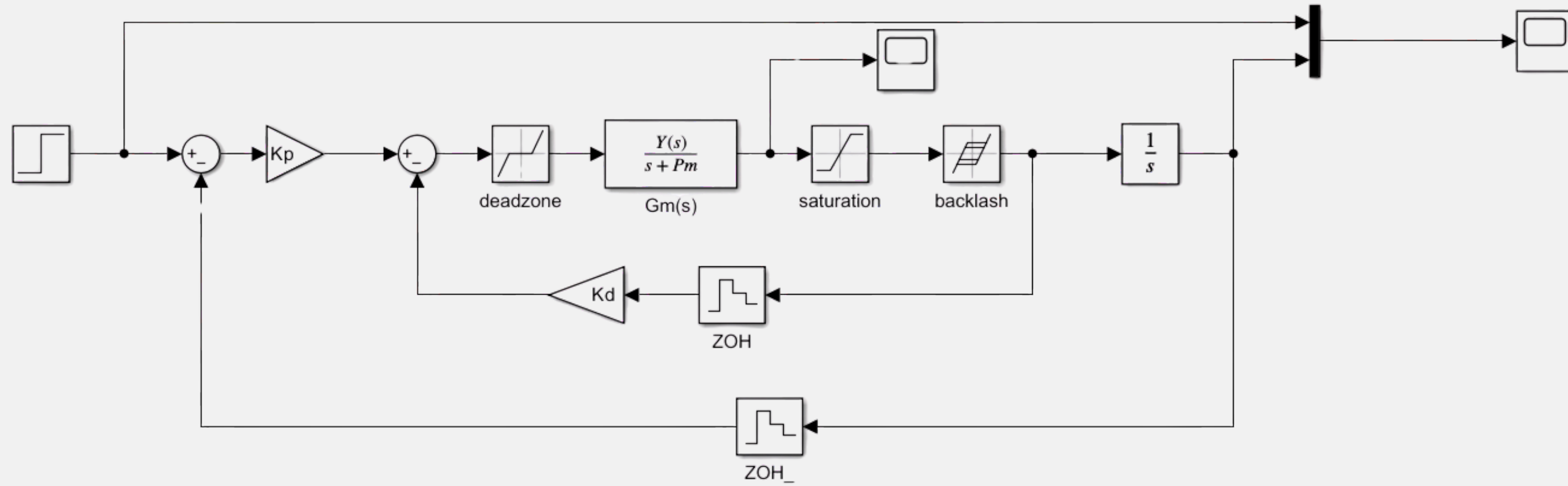
04 성능분석

설계모델



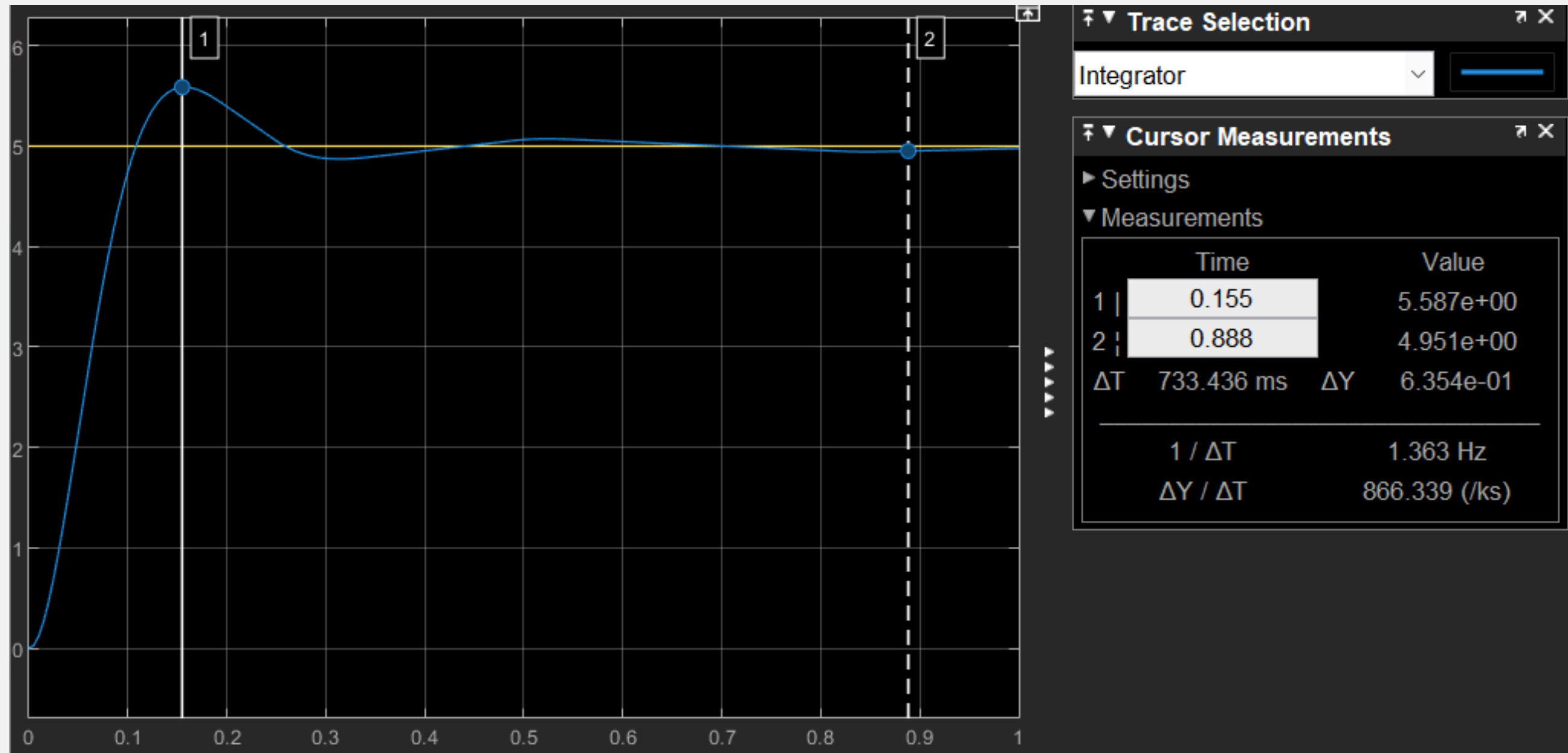
04 성능분석

해석모델 - 비선형 요소 추가



04 성능분석

해석모델

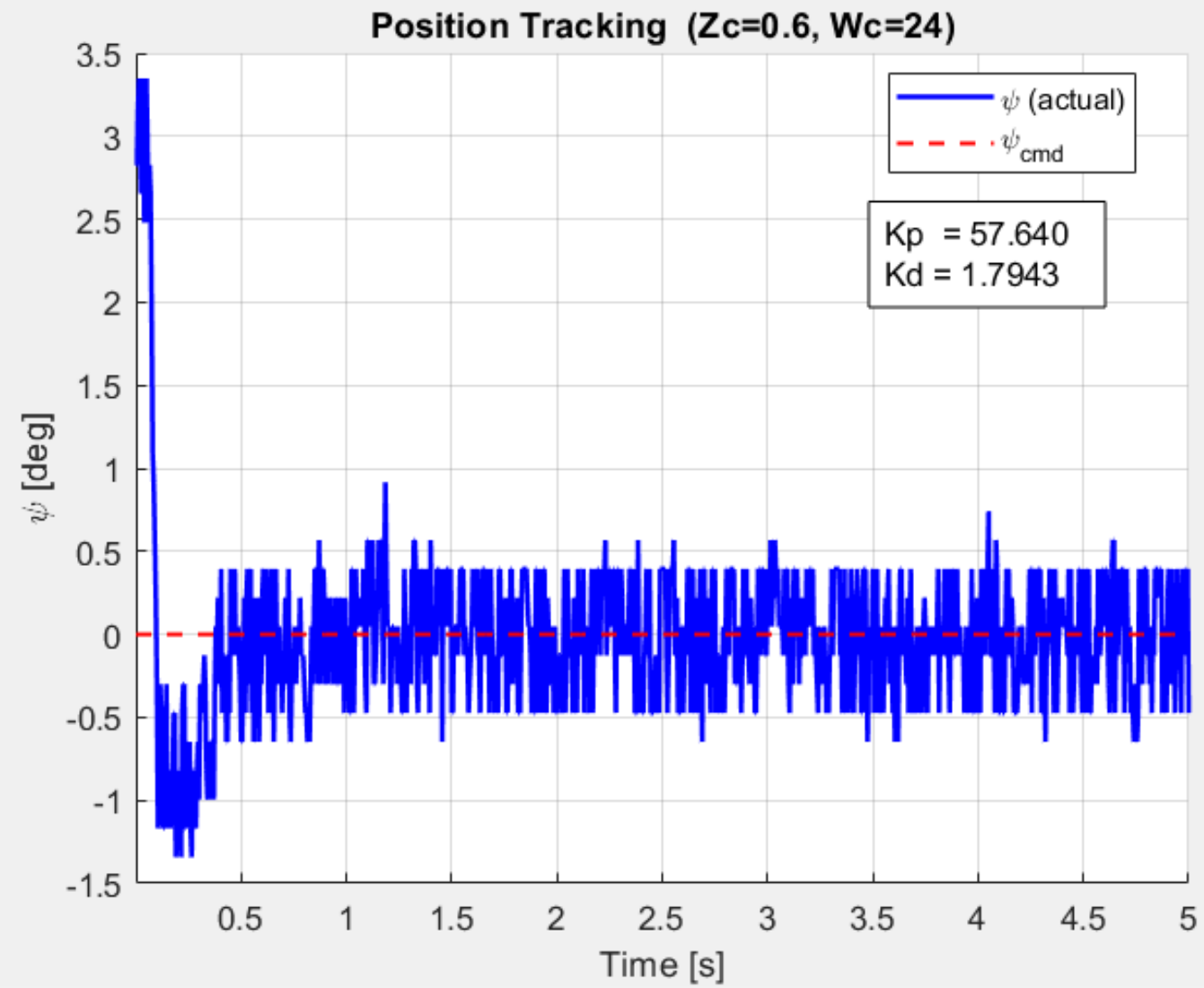


$$OS = 12.8 [\%]$$
$$tr = 0.093 [s]$$

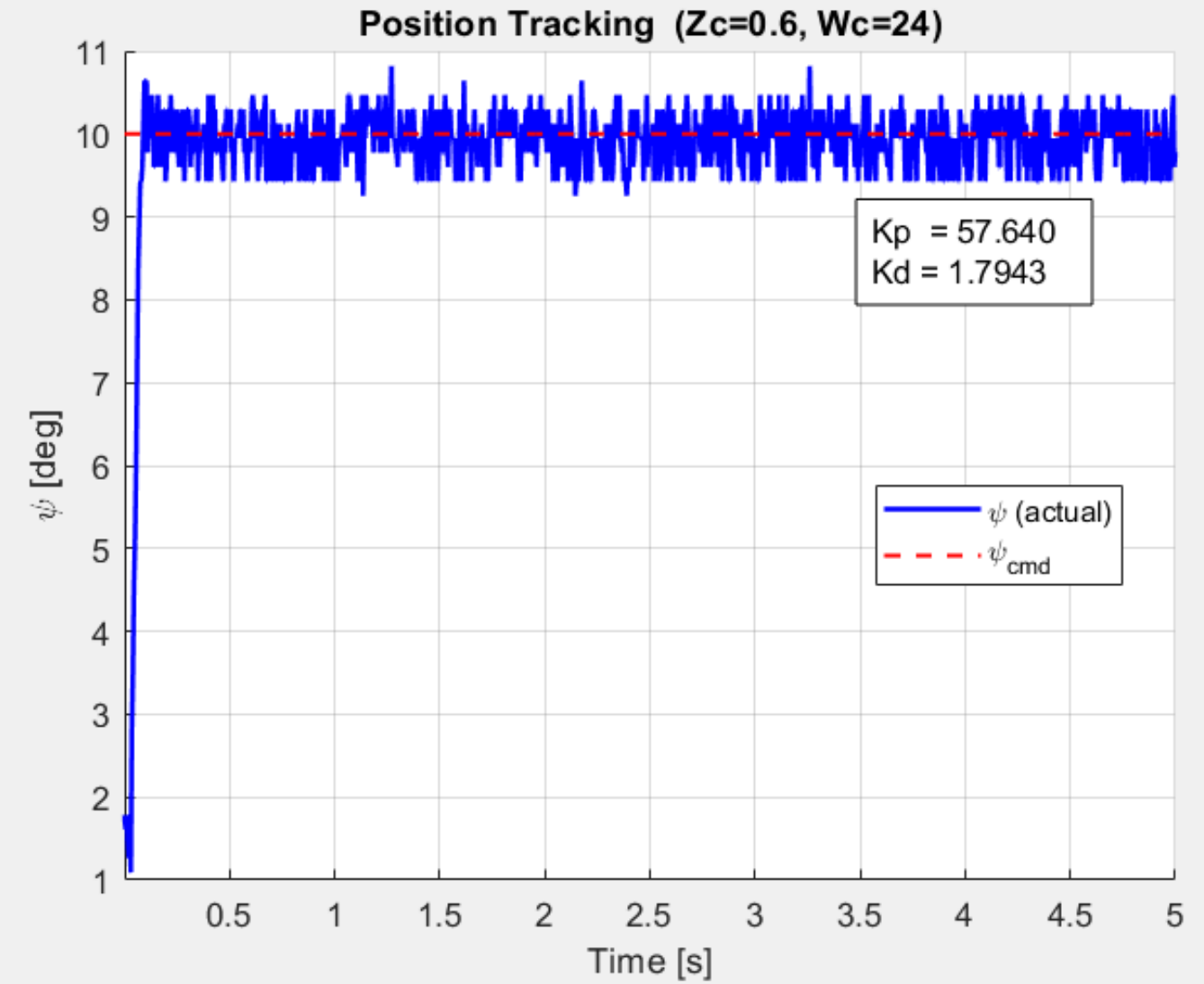
specification 초과

04 성능분석

실험결과



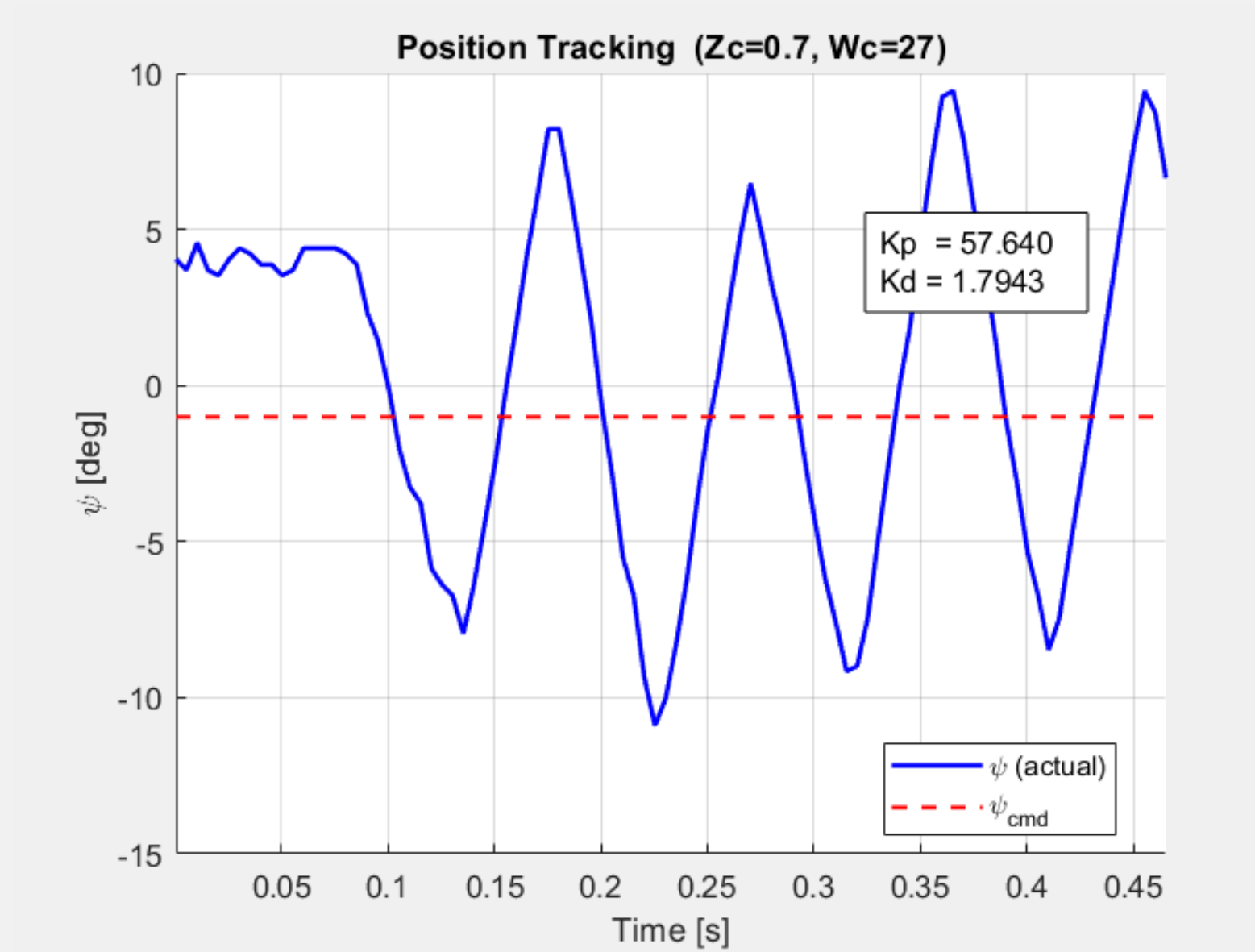
3[deg] $\rightarrow$ 0[deg]



1.7[deg] $\rightarrow$ 10[deg]

04 성능분석

실험결과



4[deg] $\rightarrow$ (-1)[deg]

Limit cycle 발생
 $\rightarrow$ omega_cmd에
Velocity saturation limit 로직 추가 예정